

SUMMARY KEYWORDS

telescope, galileo, lenses, magnification, invention, venice, jupiter, instrument, scientist
s, mountains, naked eye, astronomers, discovery, improved, impaired
vision, glass, moon, lunar cycle, looked, stars

SPEAKERS

Speaker 2 (79%), Speaker 1 (21%)

1

Speaker 1

0:00

Now turn to section four section four, you will hear a university lecture on the early development of the telescope. First, you have some time to look at questions 31 to 40? Now listen carefully and answer questions 31 to 40

2

Speaker 2

1:17

Good morning. Today's lecture is about the development of the telescope up until the end of the 17th century, we can trace the telescopes earliest form back to the 1200s. History records Roger Bacon as possibly the first to invent a magnifying glass. The principal use of his invention was for reading. This proved to be a benefit mostly for researchers who until then had been forced to give up working while still relatively young because of impaired vision. In 1608, a Dutch astronomer called Lieber Hey discovered that if convex and concave lenses were put together, they created a magnifying glass through which things at a distance could be seen. He created a miniature telescope. It was so small that initially, people took it to the theater for a better view of the stage. Galileo found out about Lieber Hey's invention in the spring of 1609, and immediately set about improving it. Galileo experimented with different lenses combining a weak convex lens with a strong concave lens, he discovered that this intensified the strength of the instrument. Unfortunately, the lenses the opticians were making for glasses were unsatisfactory for what Galileo wanted to do, as they did not have powerful enough magnification. Galileo therefore decided that to solve his problem, he would have to start the production of his own lenses. Eventually, he achieved a magnification about nine times more than you could see with the naked eye, an enormous improvement over everything else on the market. Galileo then decided to approach the Senate of Venice to demonstrate the improved instrument, many senators were eager to try the new telescope and climbed the highest bell towers in Venice to look through the glass at ships far out to sea. The government of Venice was impressed by his invention's obvious potential, and envisioned the advantages this eyeglass could have for the army. His first major astronomical discovery with the telescope was that the moon's surface was mountainous and not a perfect sphere as it had always been assumed. During the lunar cycle of 28 days, he found that the shadows seem to vary

with the light. Galileo also figured out how to estimate the altitude of the mountains on the moon by looking for bright spots in the dark areas. Galileo looked for the spots when it was a halfmoon and with geometry was able to calculate the elevation of the mountains. From these calculations, he was able to draw charts of the terrain and show the altitudes of the mountains

2

Speaker 2

4:20

in 1610, Galileo's next major discovery was regarding Jupiter using a magnification telescope 20 times the strength of his previous ones. He looked at Jupiter on successive nights. His main observation was a trio of what initially seemed to be small stars close to Jupiter. However, by the 15th night, he realized that they were going around the planet in a daily orbit, and that in fact, they were moons not stars, and three days later, he saw a fourth one. This discovery showed how important a telescope As for astronomers, they would now be able to see celestial objects that could not previously be seen by the naked eye. In the later 1600s, scientists started to make a stronger telescope, revealing the galaxy. Scientists realized that when they increased the length of the telescope, the intensity of magnification improved significantly. One such telescope was created in 1656 and magnified over 100 times. It had the incredible length of four meters. Telescopes continued to become longer over the next few years. And in fact, the astronomer have various built one in 1670, that was 43 meters long. However, these telescopes were soon to prove useless for accurate observation, because the slightest wind caused the instrument to shift. Finally, in 1675, astronomers abandoned the tube telescope and instead mounted the telescope on a building telescopes like this were called Aerial telescopes. Scientists hope that the stability of the mound would decrease the frustration when observing stars. Well, that's about it for today. And so I

1

Speaker 1

6:27

that is the end of section four. You now have half a minute to check your answers.